

JAHANVI RANA

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GitHub — Github-Portfolio — LinkedIn — Kaggle

Professional Summary

Data Analyst and BI enthusiast skilled in Python, SQL, Power BI, and ensemble machine learning algorithms, with a co-authored international research publication in Computer Vision. Experienced in building end-to-end analytics solutions, transforming raw data into actionable insights, and designing interactive KPI dashboards to solve complex business problems.

Core Competencies

Business Intelligence · Data Visualization · Predictive Analytics · Dashboard Development · Machine Learning · Data Cleaning · ETL · Data Modeling · Business Reporting · KPI Analytics · Forecasting · Statistical Analysis · Decision Support

Technical Skills

Programming & Libraries: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), SQL

Data Analytics: Exploratory Data Analysis (EDA), Data Cleaning, Validation, Feature Engineering, Time-Series Analysis, Data Storytelling

Business Intelligence: Power BI (Power Query, DAX, Data Modeling, KPI Dashboards, Drill-through, Bookmarks)

SQL Competencies: Joins, CTEs, Window Functions, Aggregate Functions, Subqueries, Views, Filtering, Grouping

Excel & Tools: Pivot Tables/Charts, Power Query, Conditional Formatting, VLOOKUP, Git, GitHub, Google Colab, Kaggle, Jupyter

Projects

OTT Streaming Analytics Dashboard

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Tools: Python · SQL · Power BI · DAX · Excel · Pandas · Matplotlib

Impact: Designed an end-to-end BI solution analyzing engagement, growth, and revenue trends for an OTT platform.

- Built automated Python (Pandas) pipelines for data cleaning, transformation, validation, and exploratory analytics.
- Designed an interactive multi-page Power BI dashboard detailing Executive KPIs, Segmentation, Geography, and Content.
- Developed dynamic DAX measures for core business metrics (Monthly Revenue, Active Users, Retention, Conversion).
- Integrated drill-throughs, bookmarks, and slicers to deliver interactive navigation and decision support.

Gold Market Intelligence & Predictive Forecasting System

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Tools: Python · Power BI · XGBoost · LightGBM · CatBoost

Impact: Built a predictive analytics platform forecasting global and Indian gold prices by merging macro indicators with ML.

- Collected, cleaned, and transformed multi-source financial datasets while engineering complex time-series features.
- Developed and compared XGBoost, LightGBM, and CatBoost ensemble models using MAE, RMSE, and R^2 metrics.
- Built Power BI dashboards visualizing Price Trends, Macro Economic Indicators, Heatmaps, and Forecasting Analysis.

Research Publication

Co-Author

2025

Co-authored research paper titled “**Enhancing YOLO with Rotated Bounding Boxes and ResNet50 Backbone for Accurate Orientation-Aware Object Detection**”, published on **SSRN (2025)**, focusing on improving orientation-aware object detection using deep learning techniques.

Education

Chandigarh University

Bachelor of Engineering (Computer Science – Artificial Intelligence & Machine Learning)

CGPA: 7.75/10

Mohali, Punjab

Expected Graduation: 2026

M M M Public School

Intermediate (Non-Medical with Computer Science)

Percentage: 88.8%

Hoshiarpur, Punjab

Passed: 2022

SGGS Sr. Sec Public School

High School / Matriculation

Percentage: 75%

Mukerian, Punjab

Passed: 2020

Certifications

- **Microsoft Learn:** Preparing Data for Analysis with Microsoft Excel
- **Microsoft Learn:** Extract, Transform and Load Data in Power BI
- **Microsoft Learn:** Data Modeling in Power BI